WHY SIGNED COMPARISONS USE NZCV FLAGS

Understanding N, Z , V flags

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UNSIGNED AND SIGNED NUMBERS

In ARM64 (AArch64) architecture, numbers are stored in memory as binary bits (0 or 1). The total number of bits allocated for a variable determines the exact mathematical boundaries of the data it can hold:

- Unsigned numbers can store a range from 0 to 2^n -1 ,where n is the number of bits.
- Signed numbers use the Two's Complement system and can store a range from $-(2)^{(n-1)}$ to 2^{n-1} -1

To represent negative numbers first bit is used as 1. To get negative value of number - flip all bites and add one.

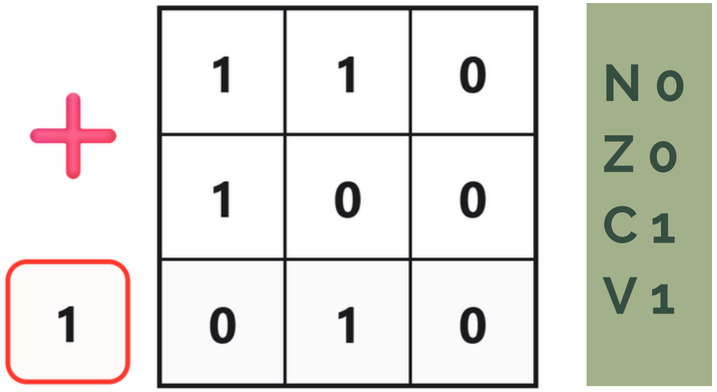
Note: for hardware optimization, the CPU performs subtraction (A - B) by calculating the two's complement of the second number and adding it (A + [-B])

Next image represents 4 bits

Bits	Signed	Unsigned
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	-8	8
1001	-7	9
1010	-6	10
1011	-5	11
1100	-4	12
1101	-3	13
1110	-2	14
1111	-1	15

FLAGS - N Z C V

- **N (Negative):** Tells if the result is a **negative** number.
- **Z (Zero):** Tells if the result is exactly **zero**.
- **C (Carry):** Tells if the math **overflowed the top** (unsigned) or needed to **borrow**.
- **V (Overflow):** Tells you if the math **broke the sign** (like adding two positive numbers, but getting a negative answer).



- Why the C Flag is ON: The math overflowed the top of the 3-bit limit. The final column produced a leftover 1 (Carry-Out) that had nowhere to go.
- Why the V Flag is ON: The math broke the sign rule because adding two negative numbers accidentally produced a positive answer.
- How the CPU detects V: The hardware calculates XOR between the Carry-In (the bit brought into the last column) and the Carry-Out (the bit pushed out of the last column). Because those two bits do not match, the V flag turns on.

To compare two numbers, the CPU performs a **subtraction.**

When you ask the computer to check if **(A > B)** or **(A < B)**, the CPU calculates **(A - B)**. It does this by using the hardware trick we discussed earlier: flipping the bits of **B** adding **1**, and then adding the result to **A.** If result is positive that means **A** is greater than **B.**

```
if (5 > 3){
    Console.WriteLine("5 is greater than 3");}
if (3 > 5){
    Console.WriteLine("3 is greater than 5");}
```

5 - 3 = 2. Two is positive, so five is greater that 3

Relationship	NZCV condition	Plain-English meaning
A < B	N ≠ V	Corrected sign says negative
A ≤ B	Z = 1 OR N ≠ V	Equal, or less than
A > B	Z = 0 AND N = V	Not equal, corrected sign positive
A ≥ B	N = V	Corrected sign says non-negative

1. Signed less than (<)

CONDITION

N ≠ V

Without overflow, N gives the sign of A − B. If overflow makes the sign misleading, V corrects the interpretation.

Example: A = −10, B = 5

−10 − 5 = −15 → 110001

N = 1 Z = 0 V = 0

N ≠ V → 1 ≠ 0 → TRUE: −10 < 5

Note: check last page for this

2. Signed less than or equal (≤)

CONDITION

Z = 1 OR N ≠ V

Z catches equality. N ≠ V catches the strictly-less-than case, including cases where overflow would mislead N alone.

Example: A = 5, B = 5

5 − 5 = 0 → 000000

N = 1 Z = 0 V = 0

Z = 1→ TRUE: 5 ≤ 5

3. Signed greater than (>)

CONDITION

Z = 0 AND N = V

Z = 0 excludes equality. N = V means the corrected sign of A − B is non-negative; together these mean strictly greater.

Example: A = 12, B = 7

12 − 7 = 5 →000101

N = 0 Z = 0 V = 0

Z = 0 AND N = V →TRUE: 12 > 7

4. Signed greater than or equal (≥)

CONDITION

N = V

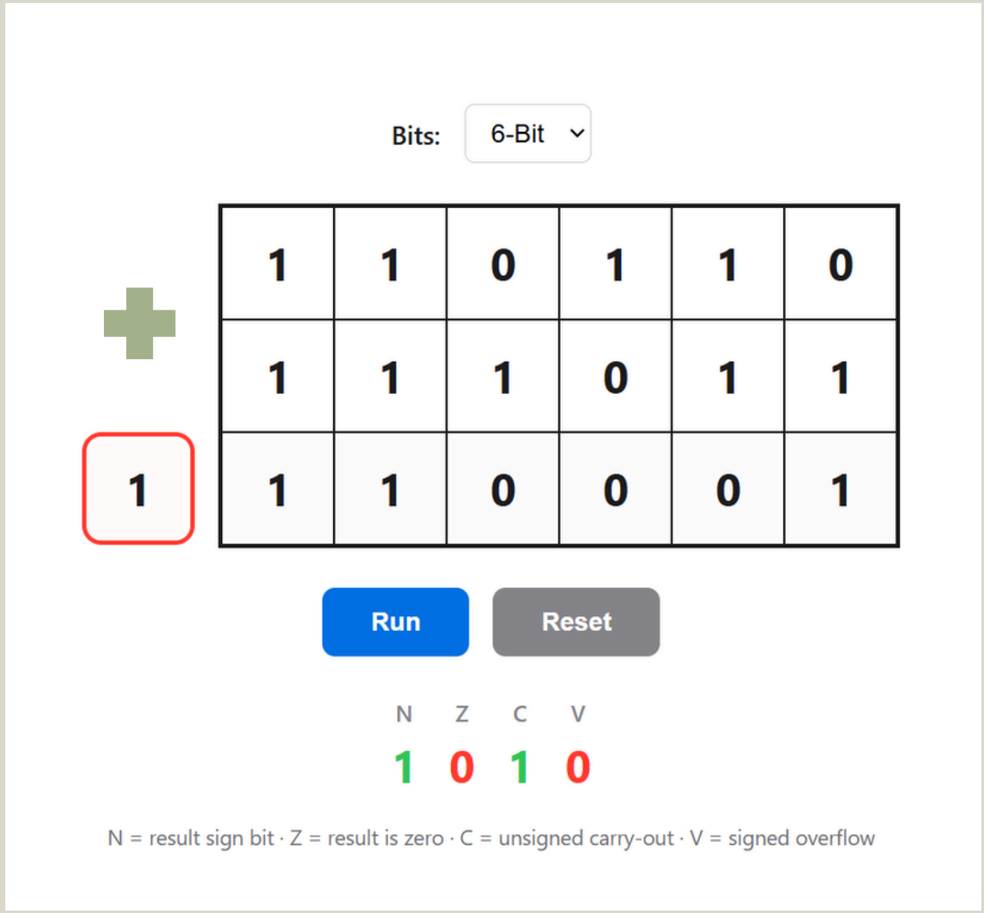
Equality is allowed. When A = B, Z becomes 1, but N = V still holds.

Example: A = −4, B = −4

−4 − (−4) = 0 →000000

N = 0 Z = 1 V = 0

N = V→ TRUE: −4 ≥ −4



Is -10 less than 5?

-10 - (-5) = 110110 + 111011 = 110001

N ≠ V → 1 ≠ 0 → TRUE: −10 < 5

AzadAshurov/**NZCV-flag-patterns-used-for-...**



1 Contributor 0 Issues 0 Stars 0 Forks

NZCV-flag-patterns-used-for-signed-comparisons/Calc.html at main · AzadAshurov/NZCV-flag-patterns-used-for-signed-comparisons

Contribute to AzadAshurov/NZCV-flag-patterns-used-for-signed-comparisons development by creating an account on GitHub.

GitHub

Click the link for NZCV Calculator